



FR 25

MA4701 AIRCRAFT DESIGN
MA4 Grp 2



FlowRider



Our Team



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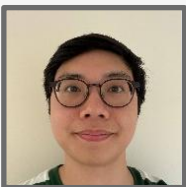
Swee Woei

Structures Lead



Jit Huat

Mission Profile Lead



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Propulsion Lead



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Materials Lead

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Cost & Feasibility



Market Analysis

Future Outlook



Sustainability

Hybrid engine that runs on SAF



Climate Change

Advanced structure and onboard avionics



Emerging Markets

Low cost and budget friendly



Increased Global Supply Chain Demands

Modular design to load cargo

Target Market



High Growth Markets



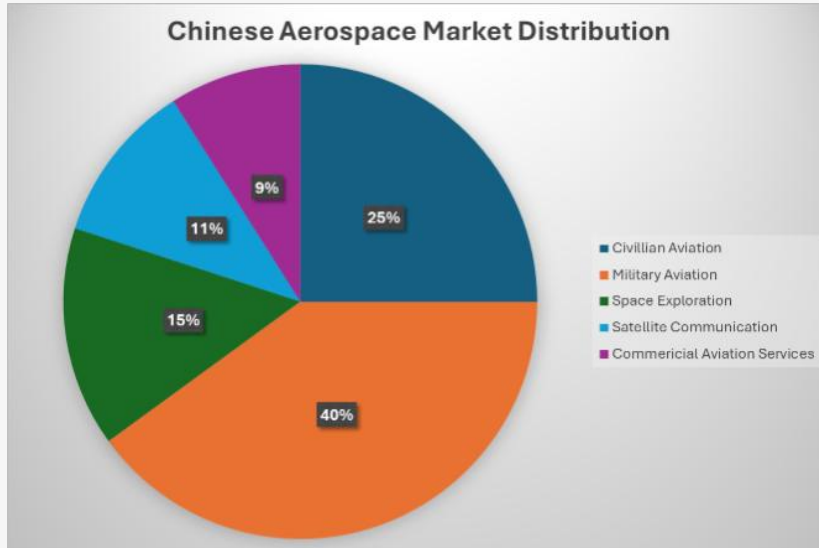
High Regional Travel
Demand



Rising Middle Income
Earners

China Market

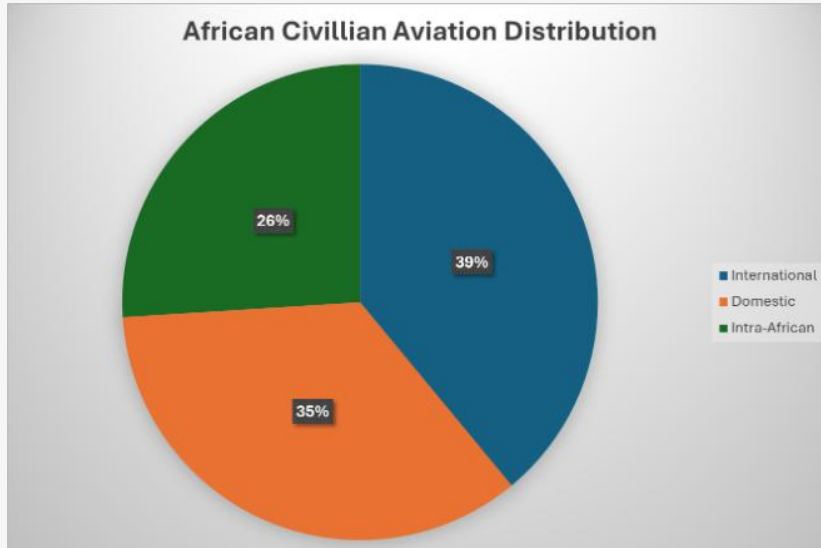
- Largest aviation sector by 2045 - US\$61bn
- **25% Civilian Aviation**
- Possible to capture **17%** of market share



Aircraft Model(s)	Manufacturer	Seat Capacity	Commerical Cargo Capacity	Market Share
C919	COMAC	156-168 Passengers	NA	20.8%
B737	Boeing	138-230 Passengers	NA	32.0%
A320	Airbus	140-170 Passengers	NA	30.2%
FR25	FlowRider	48-88 Passengers	0 Kg with 88 Passengers 4800Kg with 48 Passengers	17 %

Africa Market

- Annual growth rate of 9.9%
- **61% Domestic Flights**
- Possibly capture **20%** of domestic market share



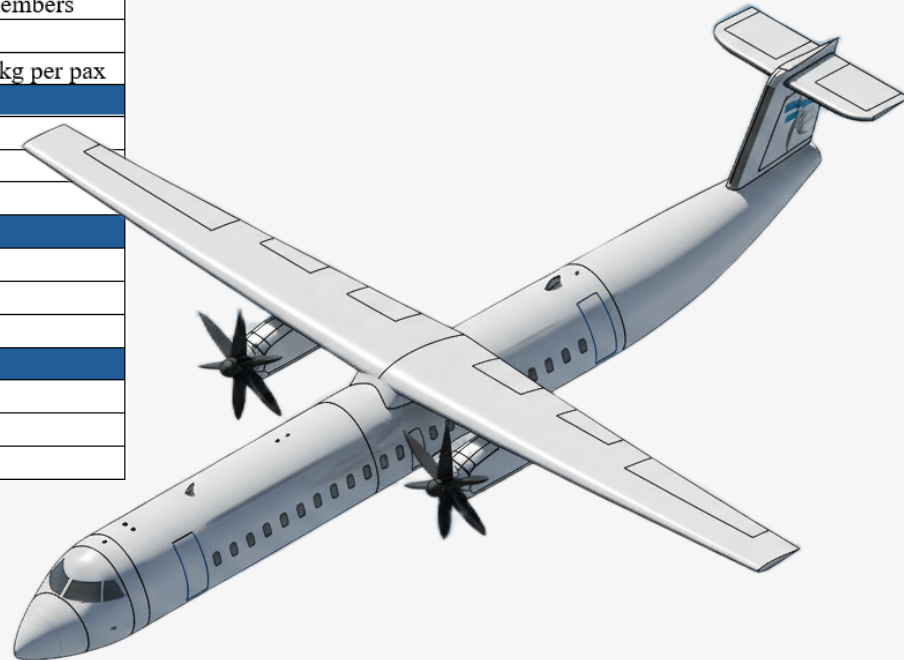
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C919	COMAC	156-168 Passengers	NA	20%
B737	Boeing	138-230 Passengers	NA	20%
A320	Airbus	140-170 Passengers	NA	20%
D328eco	Deutsche Aircraft	40-50 Passengers	NA	20%
FR25	FlowRider	48-88 Passengers	0 Kg with 88 Passengers - 4800Kg with 48 Passengers	20%

A silver twin-engine turboprop aircraft is shown from a low angle, moving along a runway. The aircraft has two large propellers and a high-wing configuration. The background features a green field and a blue sky with light clouds. A blue circular graphic element is visible on the right side of the image.

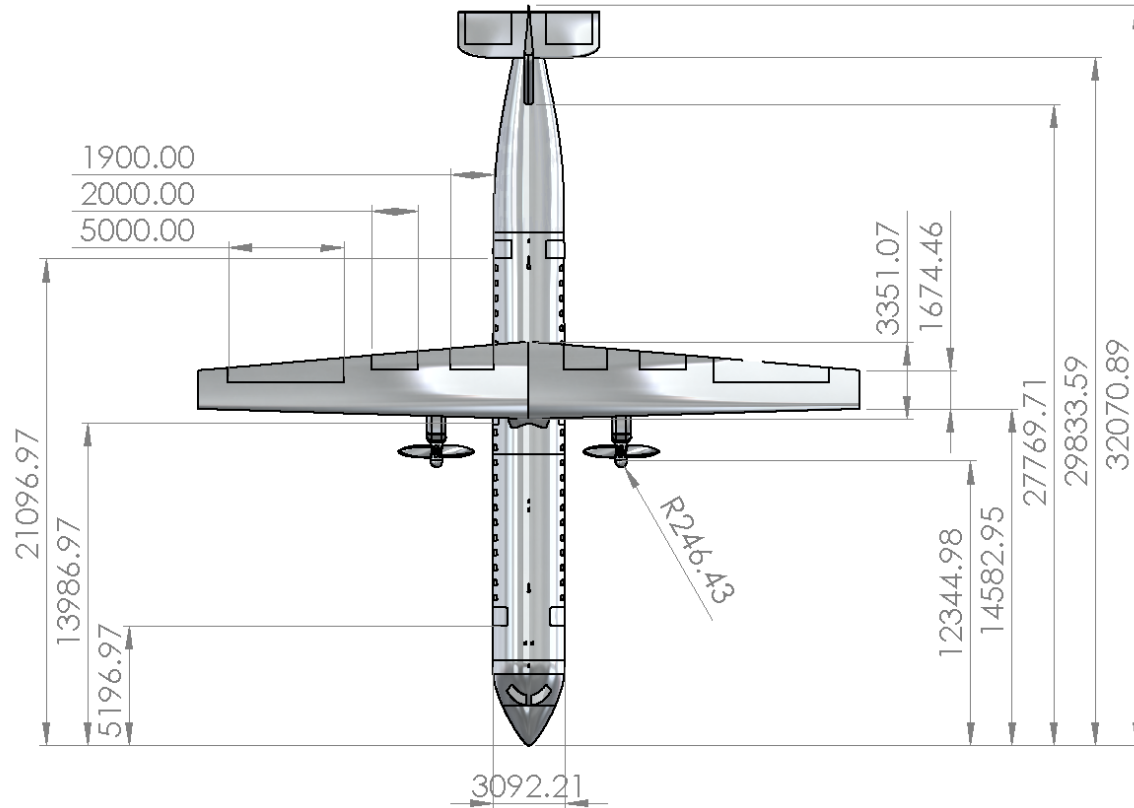
Aircraft Overview

Key Specifications

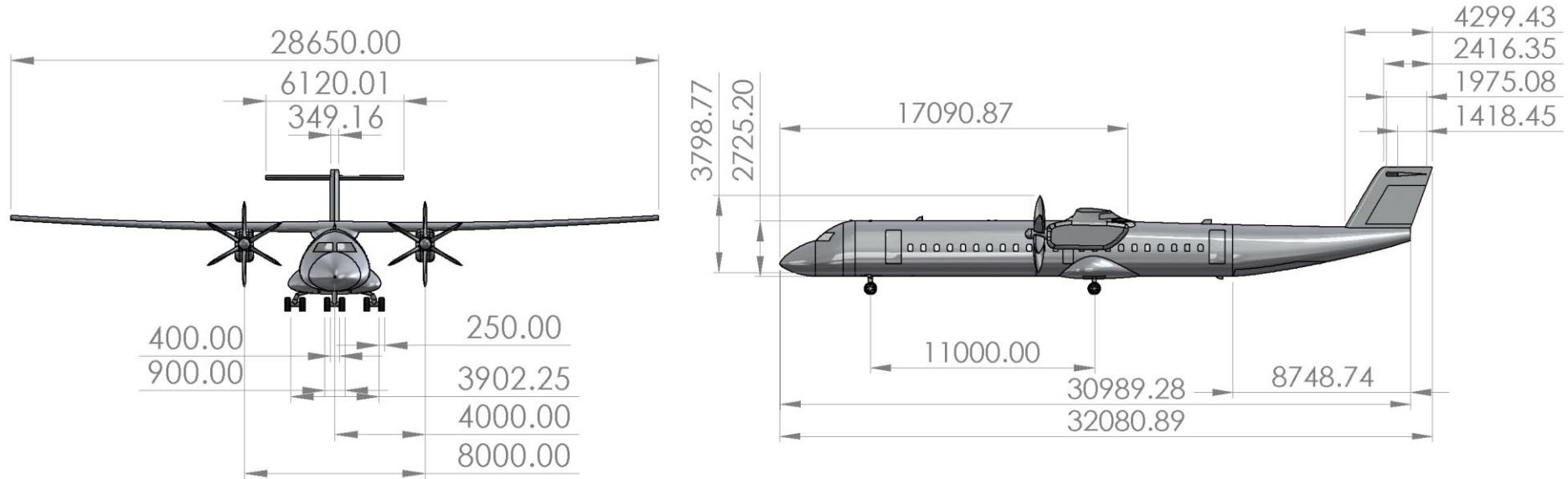
Basic Information		
Number of Passengers (pax)	88	4 Crew Members
Range (Max Payload, Km)	1500	
Payload (kg)	9200	Assume 100 kg per pax
Dimensions (m)		
Length of Fuselage (m)	33.04	
Diameter of Fuselage (m)	3.09	
Wingspan (m)	28.65	
Weight Data (kg)		
MTOW	30674.3	
Empty Weight	19615	
Fuel Weight	1859.3	
Performance Parameters		
Cruise Speed (knots)	360	
Take-off Runway (m)	1700	
Landing Runway (m)	1070	



General Dimensions



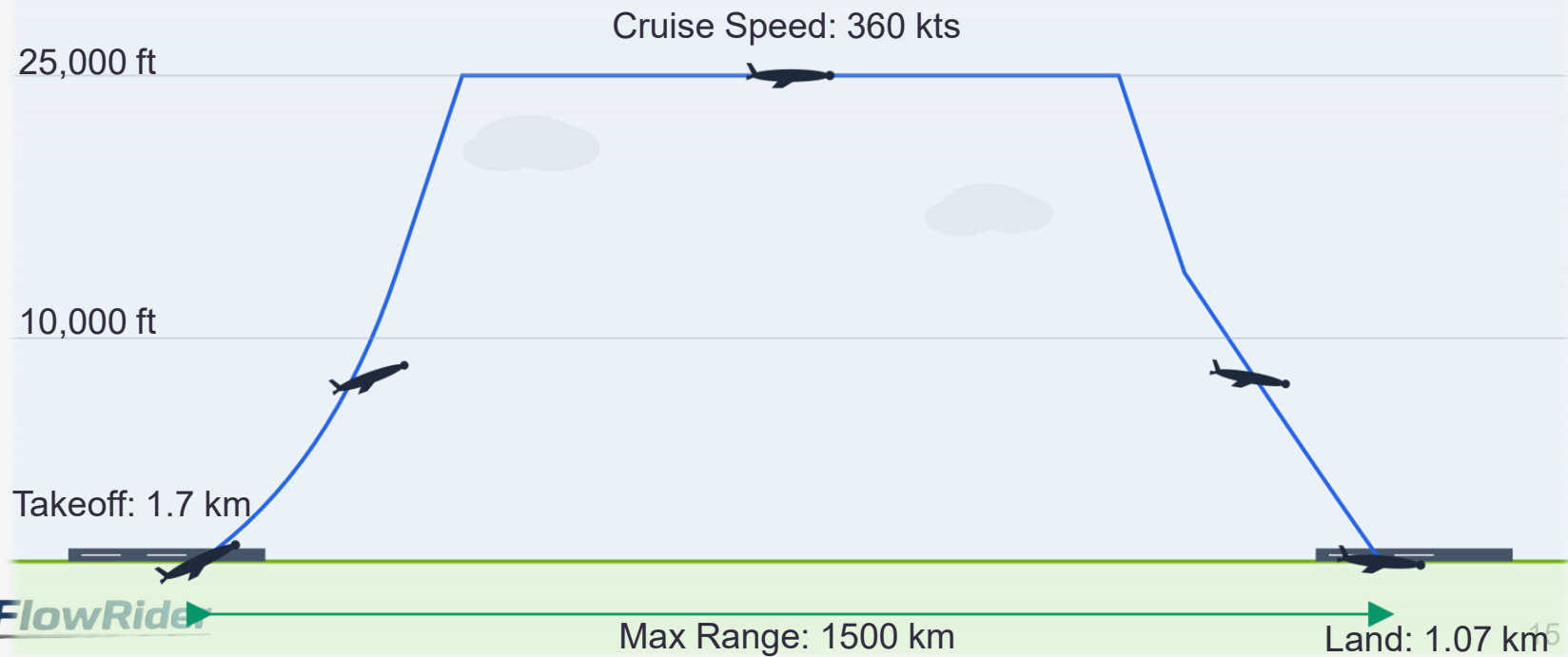
General Dimensions



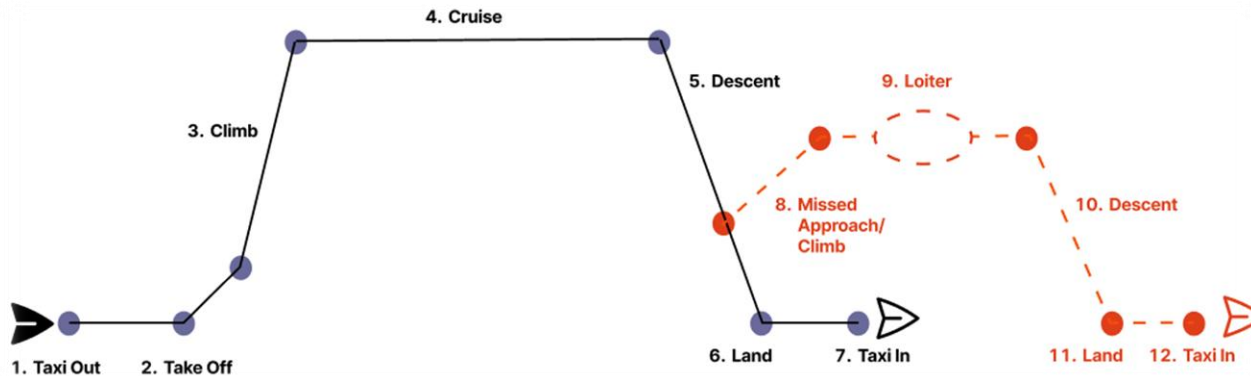


Mission Profile

Mission Description



Mission Profile



1500 km

9200 kg payload

92 (88+4) Pax



Design Overview

What's Special?



Commercial Regional Aircraft

- Only one configuration (Passenger / Cargo)



FR25

- Modular Cabin Configuration
- 88 Passenger Seats
- 48 Passenger Seats + 4000 kg Cargo

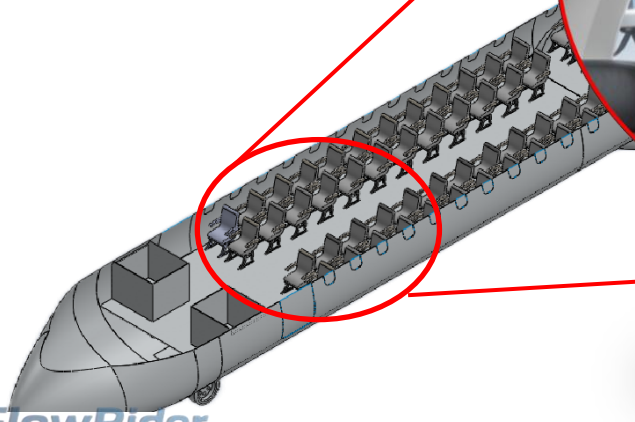
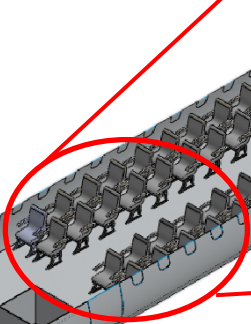
Full Assembly



Passenger Configuration

88 passengers

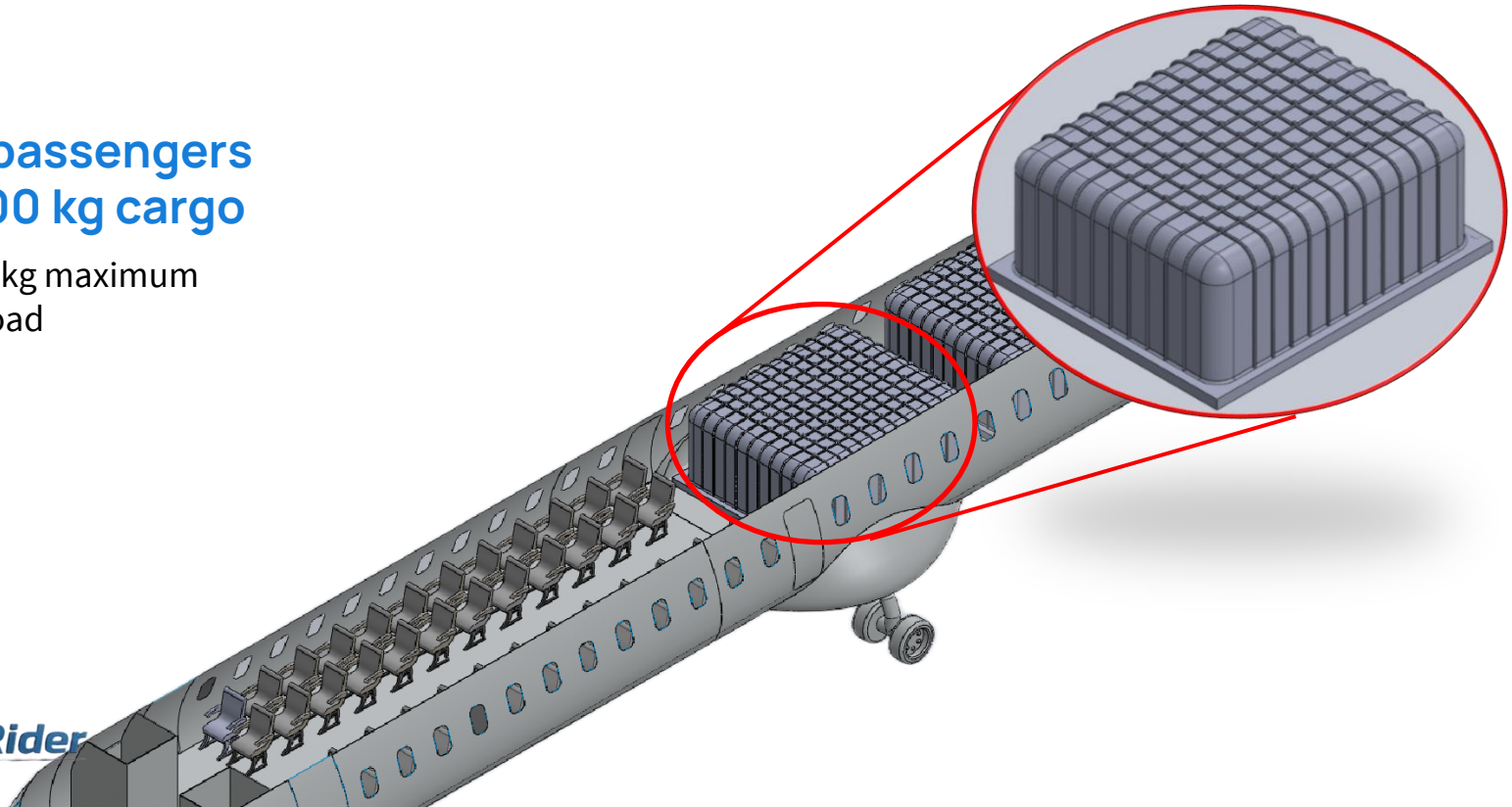
9200 kg maximum
payload



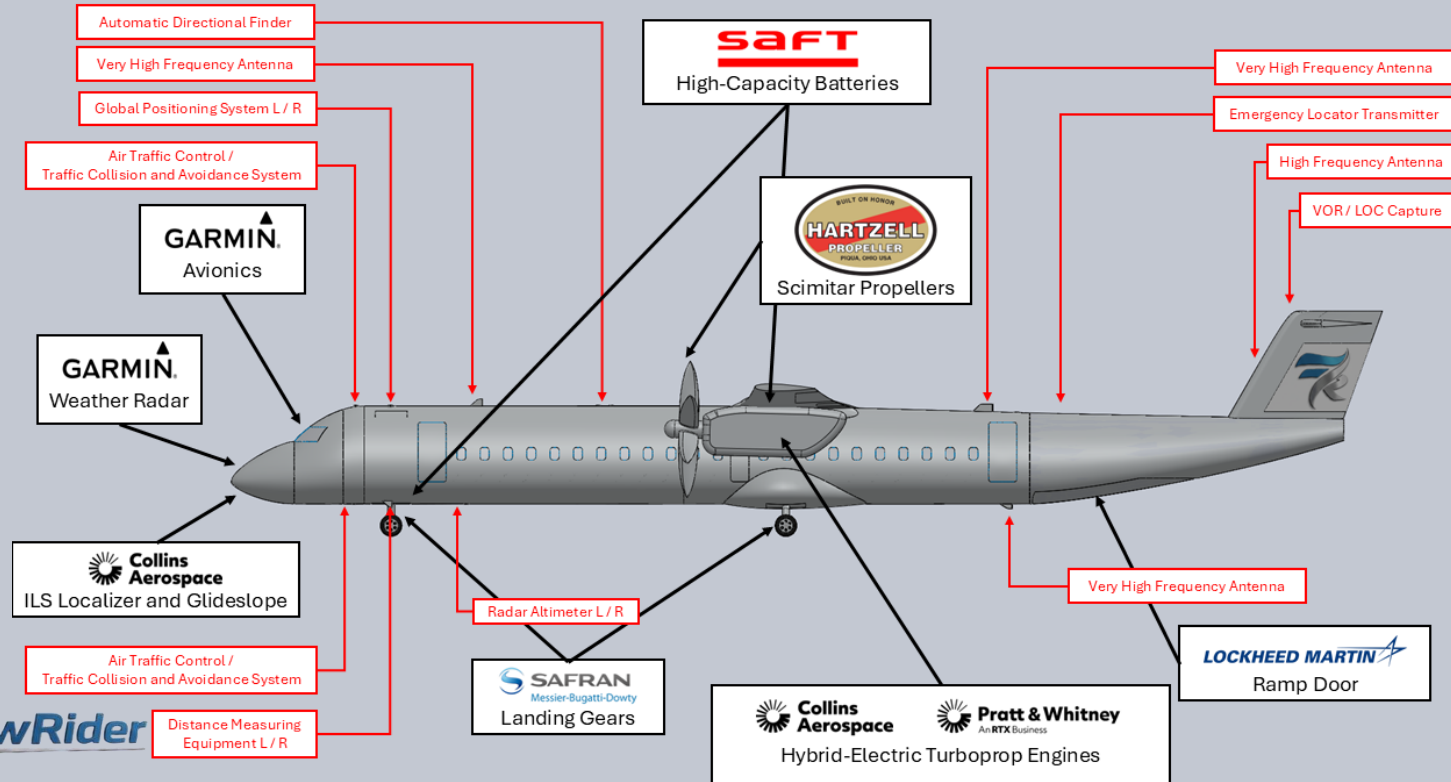
Modular Configuration

48 passengers
4000 kg cargo

9200 kg maximum
payload



Component Sourcing



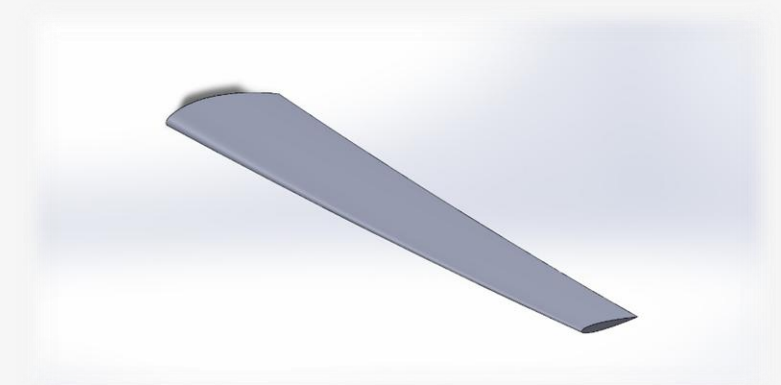


Wing Design

Wing Design

● Purpose

- Provide sufficient lift to carry out mission (passenger + range)
- Maintain low speed performance
- Ensure sufficient space for installations (hybrid system + fuel)
- Retain structural integrity

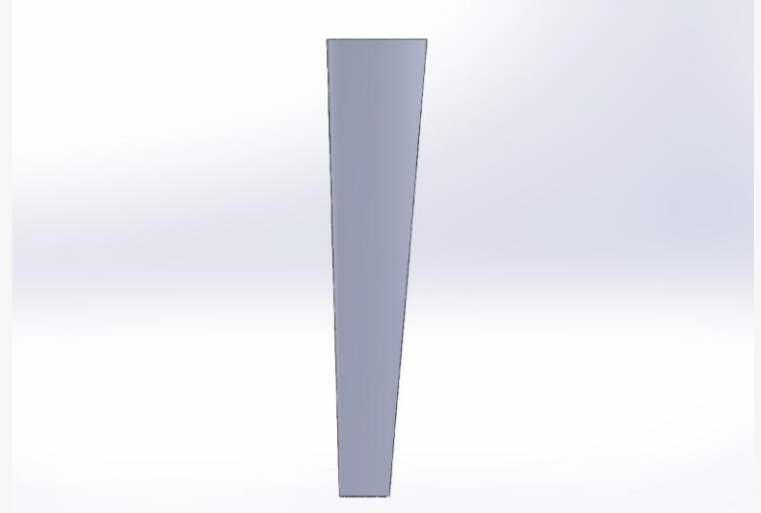


Isometric View of Half-Wing

Wing Design

● Guiding Principles

- Cruise at $M_{0.4} - 0.5$
- Short field operations → low speed, high lift
- Propeller slipstream → thicker, cambered root profile
- Deals with adverse weather conditions

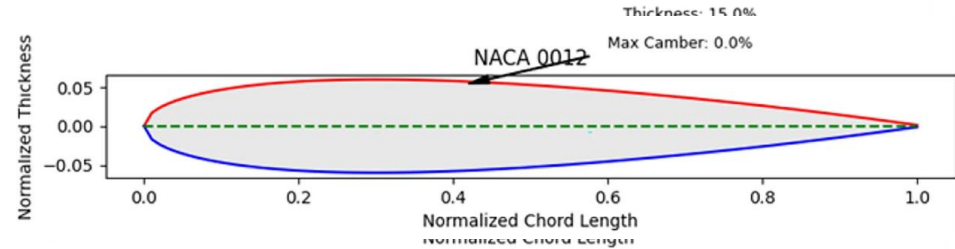


Top View of Half-Wing

Airfoil Selection

● 3-profile

- Smooth taper transition
- Better lift distribution & stall characteristics



Position	Airfoil	Function
Root	NACA 4415	High lift, structural depth, mild stall
Mid	NACA 2412	Smooth transition, balanced camber
Tip	NACA 0012	Low drag, docile stall

Main Wing Geometry

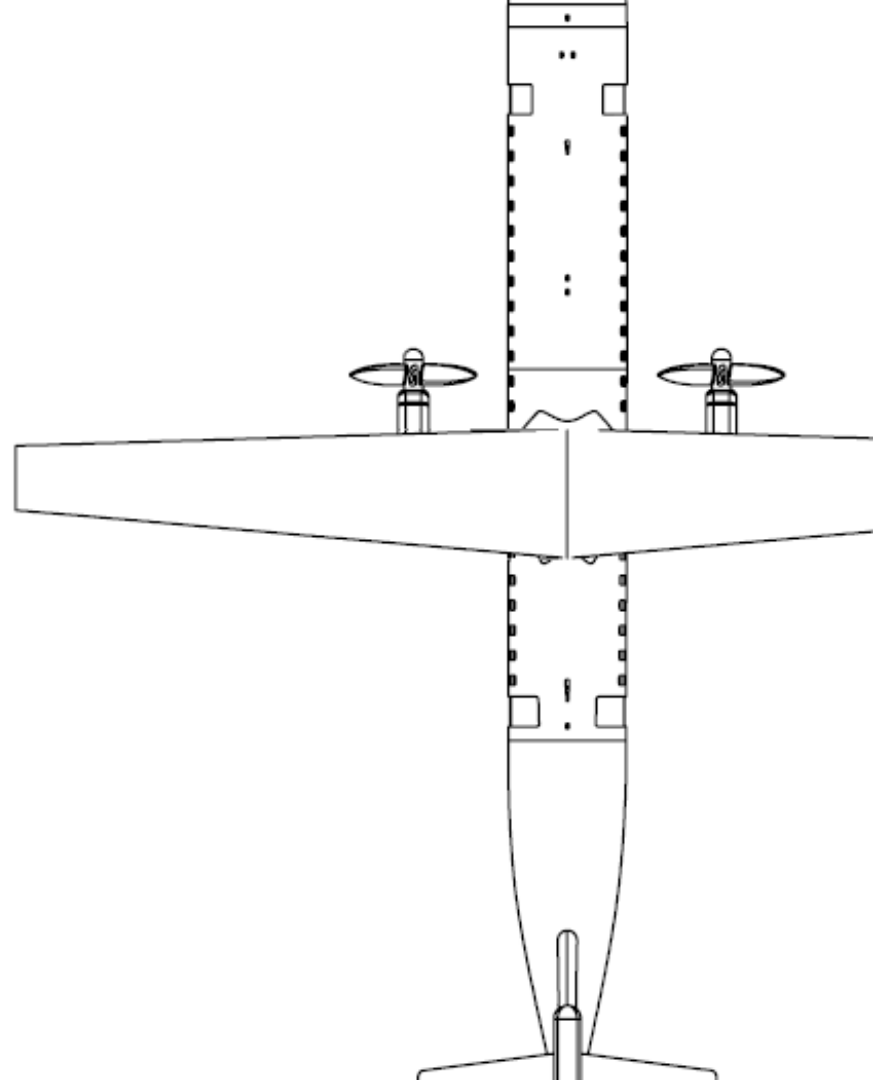
Wing Loading (W/S)	4.2 kPa
Wing Area (S)	72 m ²
Aspect Ratio (AR)	11.4
Wingspan (b)	28.65 m
Taper Ratio (λ)	0.5
Root Chord (c_r)	3.351 m
Mid Chord ($c_{0.5}$)	2.513 m
Tip Chord (c_t)	1.675 m
Sweep Angle (Λ)	0° (aligned at 0.25c)



Wing Features

Straight wing

- Swept-back usually for high-speed operations
- Offers high lift at low speed
- Simple structure
- Better roll response



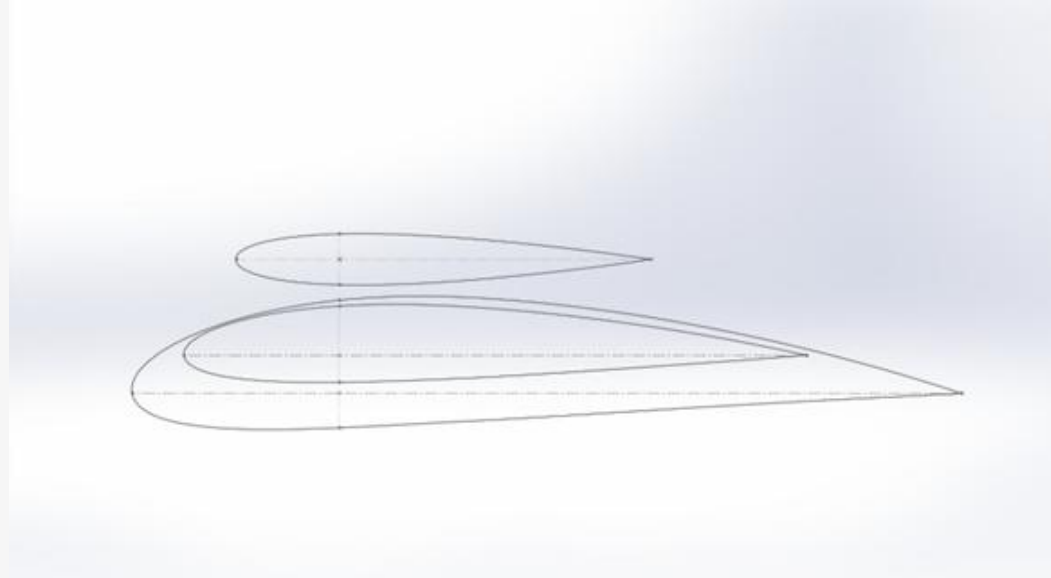
Wing Features

Dihedral

- Polyhedral adoption
- Effective angle $\approx 2.1^\circ$
- 1° (mid) $\rightarrow 3^\circ$ (tip)

Washout

- 2.5° (root to tip)
- Promotes root first stall
- Maintains aileron control



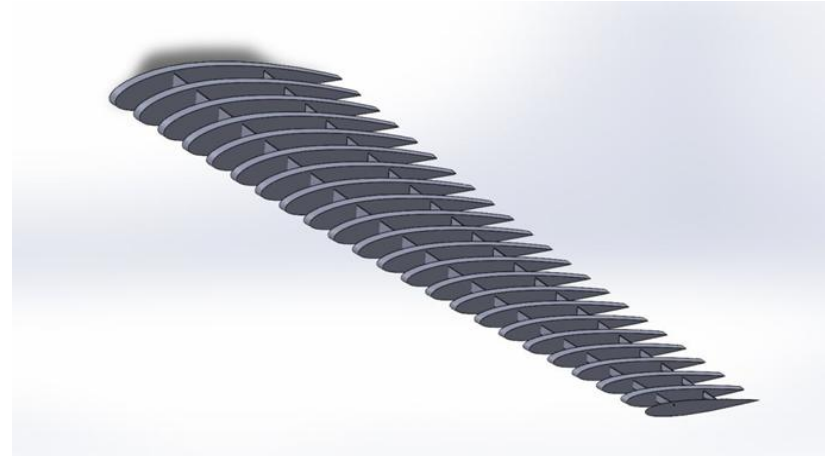
Structural Design

2-spar torsion box configuration

- Front spar at 25% chord
- Rear spar at 65% chord
- Wing skin & Stringers complete torsion box

Load Path

- Lift → Skin & Ribs → Spars → Fuselage



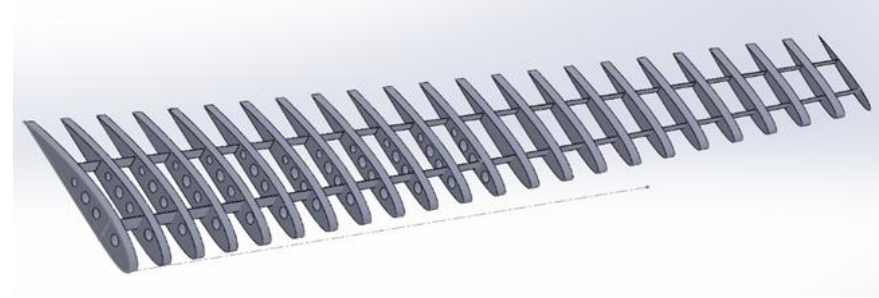
Fuel Integration

Fuel Mass

- 1675 kg $\approx$ 2.1 m³ (Sustainable Aviation Fuel)

Integral Fuel Tanks

- Stored within wing box
- Spanwise tank length (max): 10 m
- $\sim$ 2.3 m³ volume available @ $\sim$ 70% volumetric efficiency



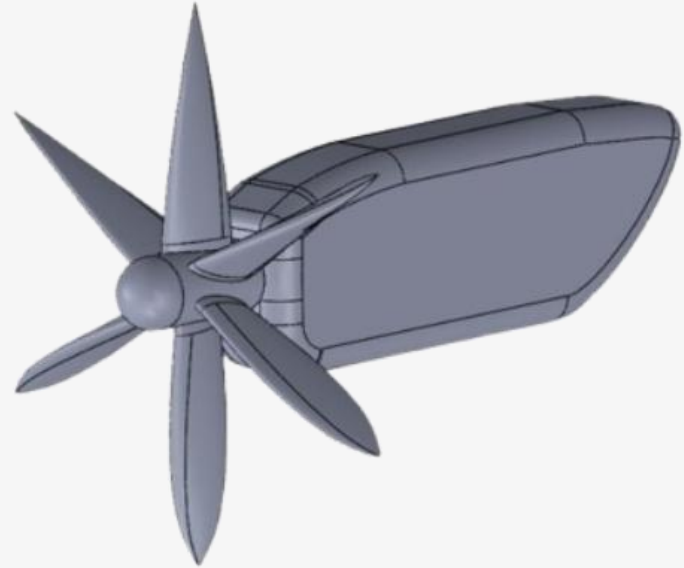


Propulsion

Engine Selection

● Hybrid Electric Engine

- Reduced fuel consumption and overall emission
- Shorter take off distance
- Assist take off and climb phase
- Reduced noise emission



Engine Performance

2,492 kW

Mechanical Power



890 kW

Electrical Power



3,382 kW

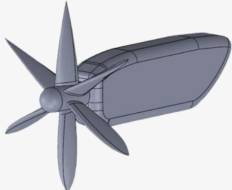
Combined Power

6,764 kW

Total Propulsive Power



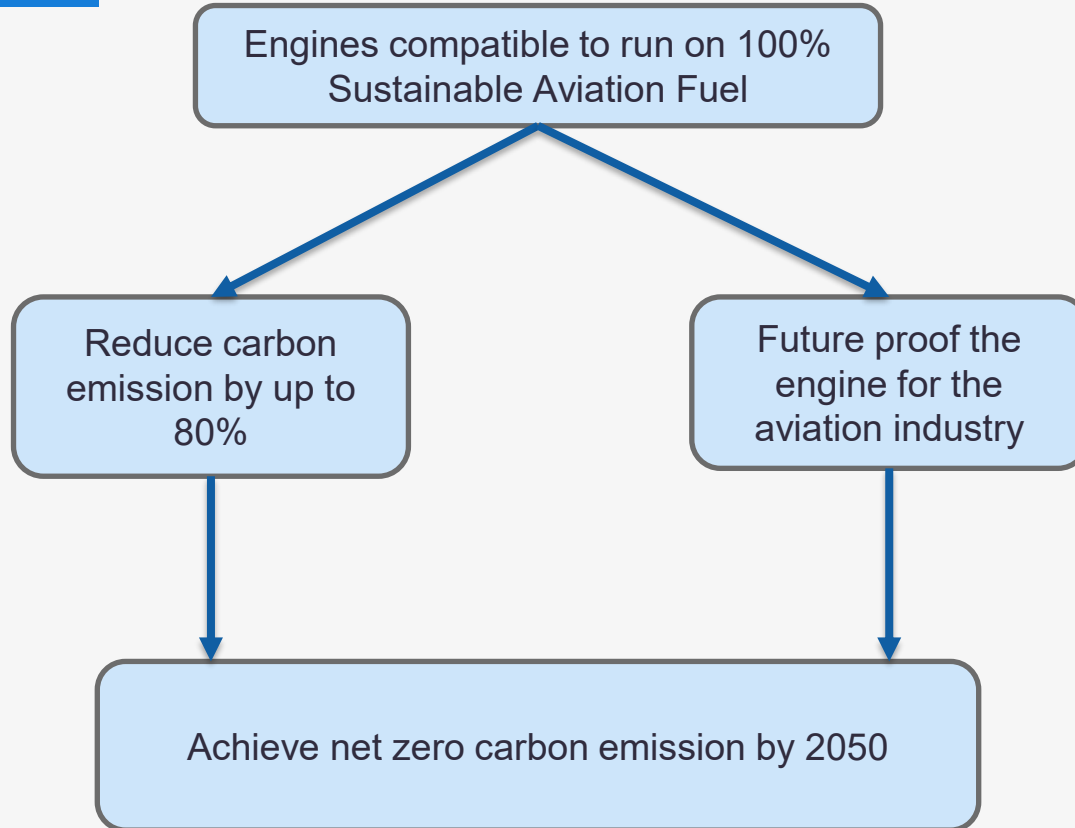
Fuel Efficiency

	Pratt & Whitney PW100	Pratt & Whitney PW150A	Pratt & Whitney PW38HE
Engine			
Power (kW)	4,045	7,562	6,764
PSFC (kg/kWh)	0.23	0.24	0.171

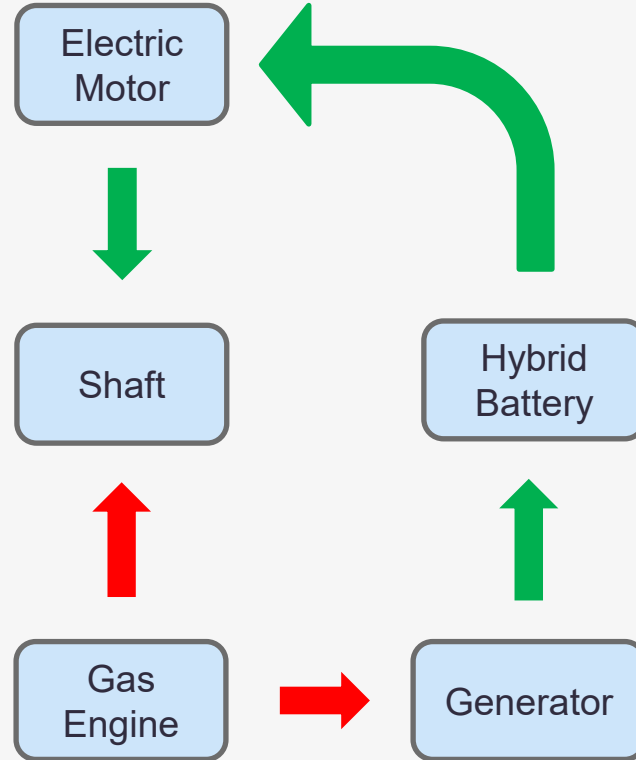
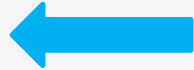
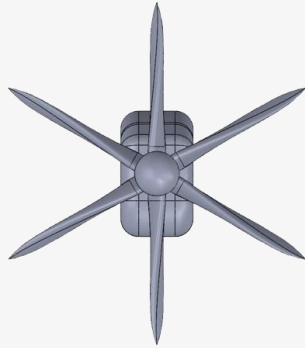
FR25 Hybrid Electric Engines can achieve comparable power output with **20% to 30% lower** in fuel consumption



Emission



Hybrid Operation



Battery Safety

● Mitigating Risk of Battery Fire

- Reinforced housing enclosure to protect battery from impact
- Coolant device fitted to housing to regulate temperature
- Thermal barrier between battery modules to prevent heat transfer





Material Selection

Material Selection

What's Different?

- 60-70% usage of composite materials, exceeding regional competitors and approaches widebody standards
- Negligible / Zero Corrosion compared Dash 8 running in coastal humidity
- 24-28% weight reduction (for fuselage and wings)

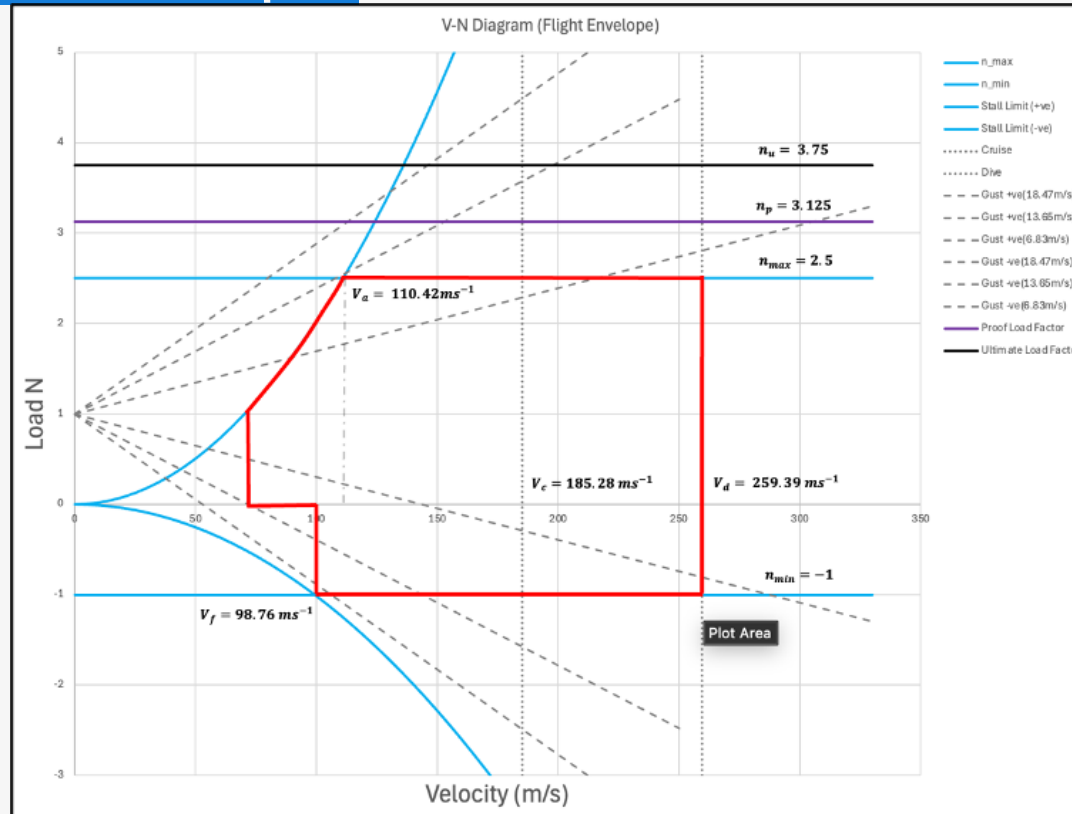
Component	Materials	Key Benefits
Fuselage	CFRP (60%) + Al-Li (35%) + Ti (5%)	Fatigue resistance, no corrosion
Wing	CFRP (70%) + Al-Li (25%) + Ti (5%)	High strength-to-weight ratio
Tail	CFRP (65%) + Al-Li (30%) + Ti (5%)	Stiffness, corrosion resistance
Interior	Aramid honeycomb, Al-Li frames	Lightweight, FAR 25.853 compliant

Aircraft	Composite Usage	Primary Materials	Structural Philosophy
FR25 (Our Design)	60-70%	CFRP + Al-Li 2050 + Ti	Hybrid optimization
ATR 72-600	30%	CFRP + Al alloys	Partial composite wing
Dash 8 Q400	5%	Aluminium alloys	Primarily metallic
Boeing 787	50%	CFRP + Ti	All-composite fuselage
Airbus A350	53%	CFRP + Ti + Al-Li	Advanced composites

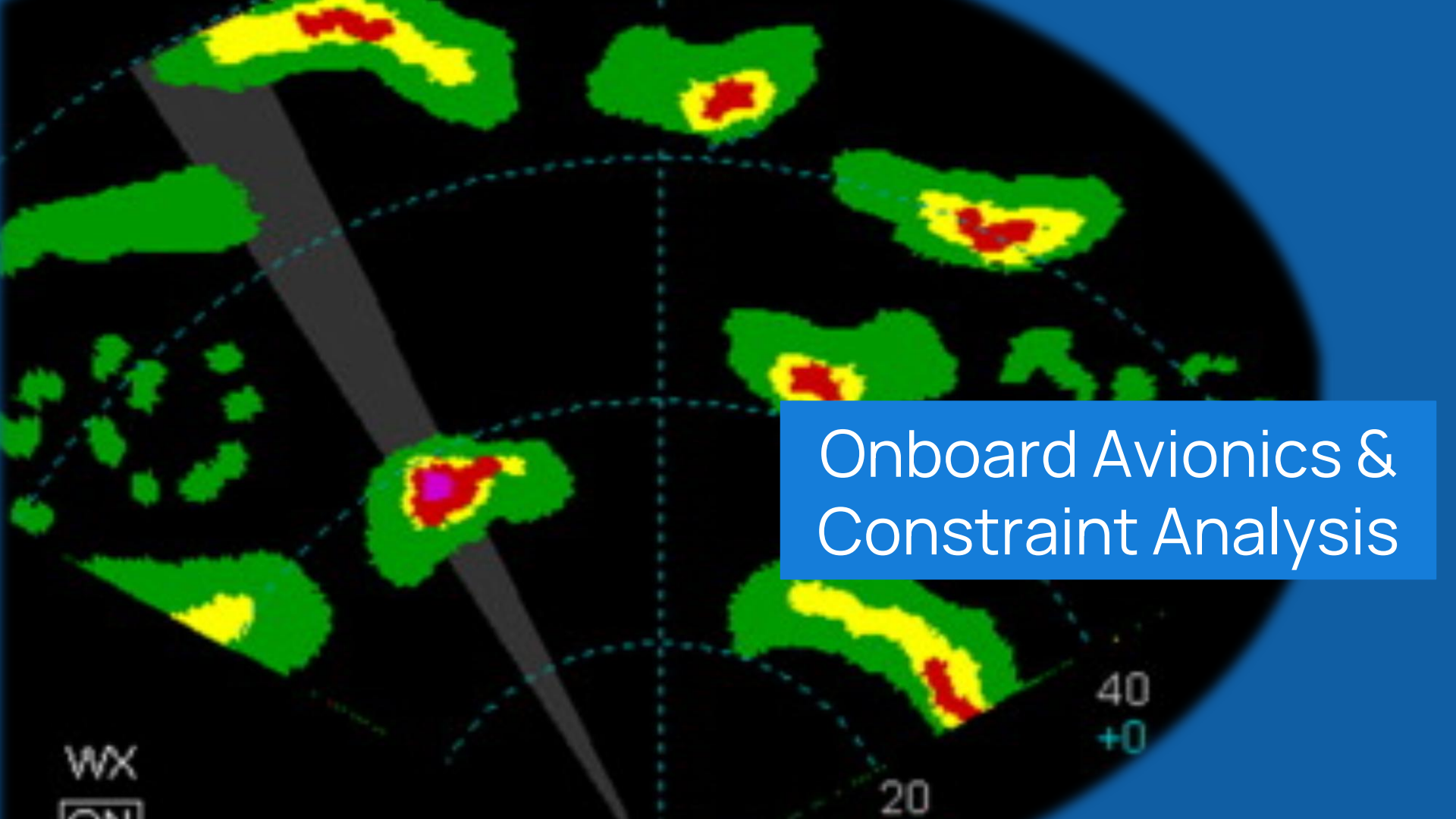
A detailed view of a modern aircraft cockpit. The instrument panel features multiple large digital displays showing various flight parameters, including altitude, speed, and engine status. The pilot's seat is visible on the left, and the center console with control yokes and throttle levers is in the foreground. The large windshield provides a clear view of a bright blue sky with scattered white clouds. A blue semi-transparent banner is overlaid on the right side of the image.

Flight Envelope

Flight Envelope



Onboard Avionics & Constraint Analysis



WX

ON

40
+0

20

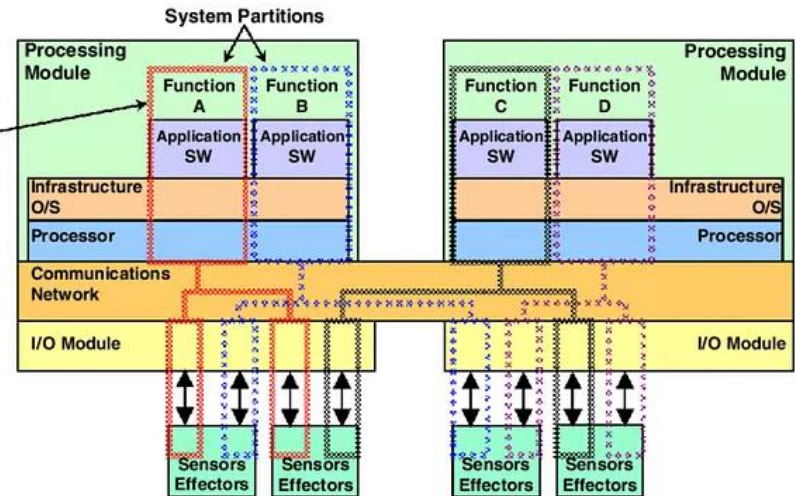
Avionics System Architecture

● Integrated Modular Avionics (IMA)

- Reduced weight compared to traditional system architectures
- Consists of 4 CPMs for redundancy

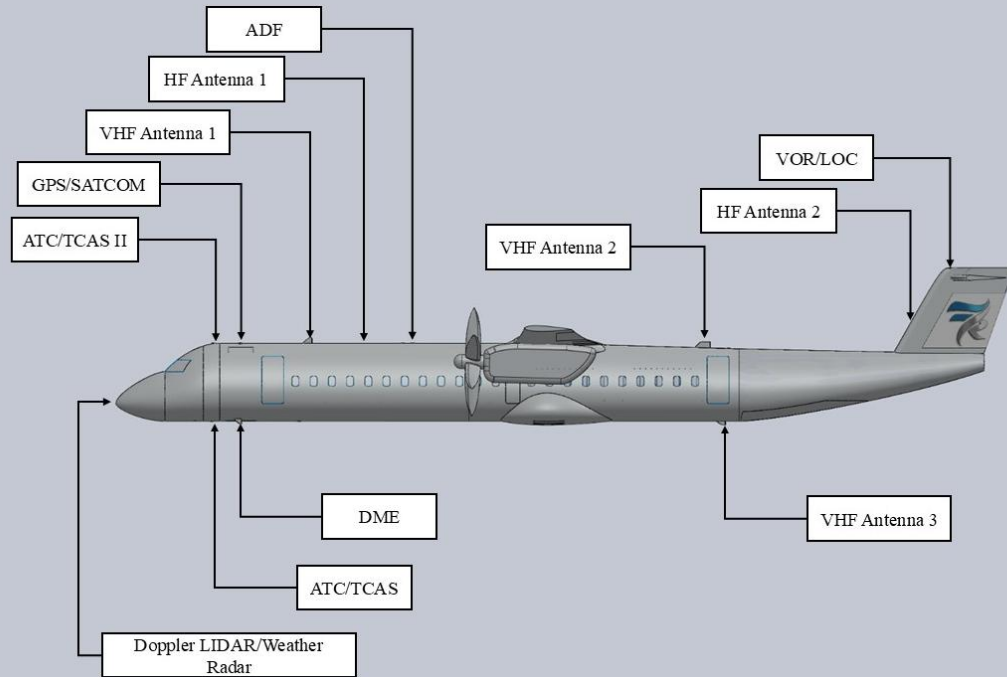
**IMA
System
Architecture:**

Virtual System
Boundaries



Communication System

● Comprehensive Connectivity



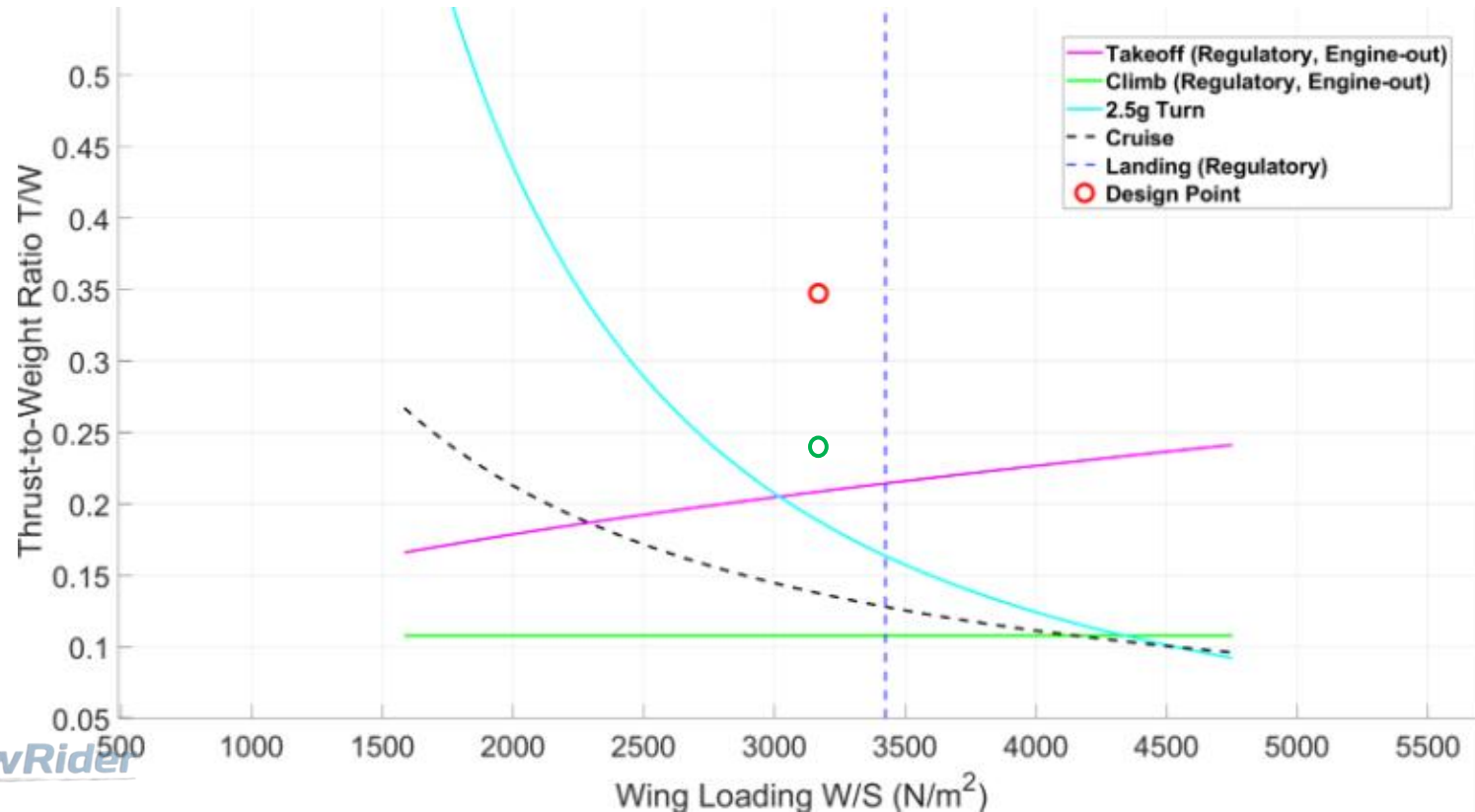
Safety Systems

● Advanced Turbulence Detection and Mitigation System

- Forward-looking LIDAR system emits laser pulses to detect variation in air velocity, indicating turbulence
- Active Gust Load Alleviation System (GLAS) that reduced turbulence effects by up to 80%

Feature	Specifications
Detection Range	14 – 17.5 km
Laser Output	3.3W Continuous Operation
Beam Diameter	150mm + 50mm dual-beam configuration
Weight	84 kg
Update Rate	1 Hz

Constraint Analysis

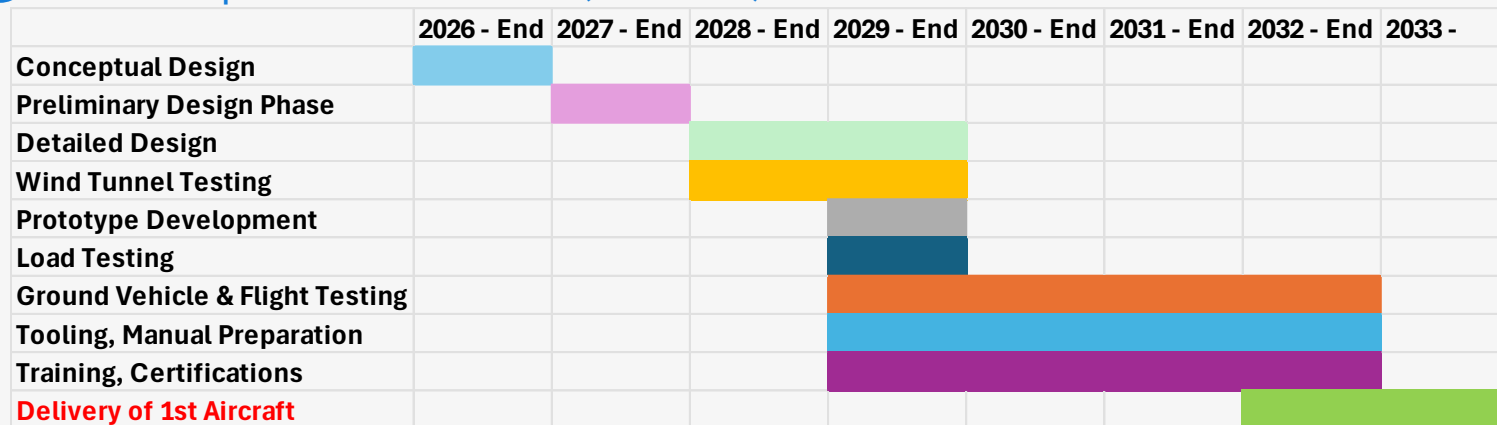




Cost & Feasibility

FR25 Program

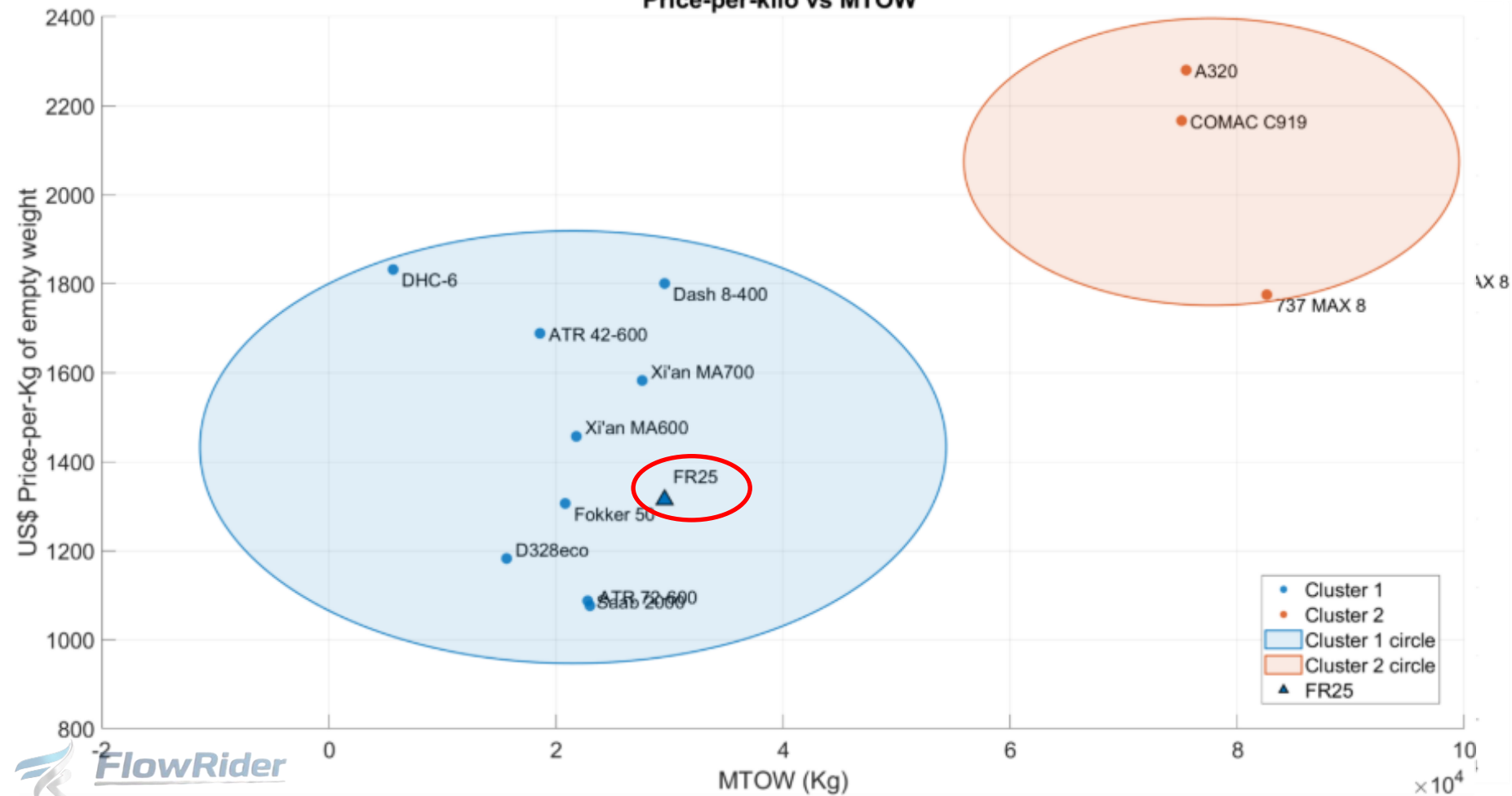
Development Timeline (7 Years)



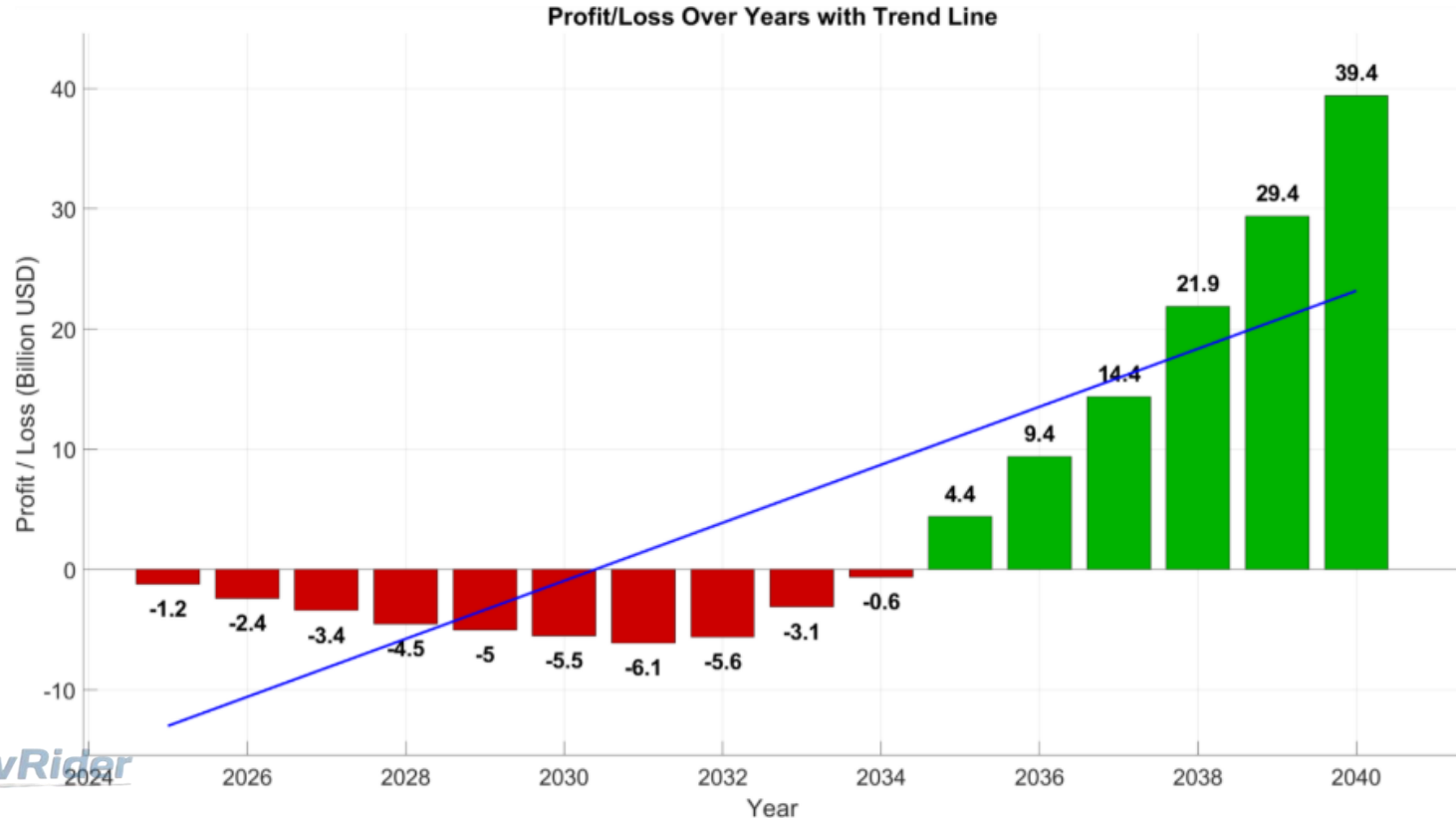
Program Cost

- **US\$6.1 billion**
- First delivery from **Year 7**

Price-per-kilo vs MTOW

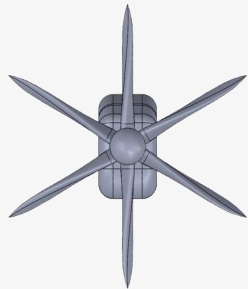


Breakeven Analysis



Conclusion

Hybrid Propulsion Engine that runs on SAF

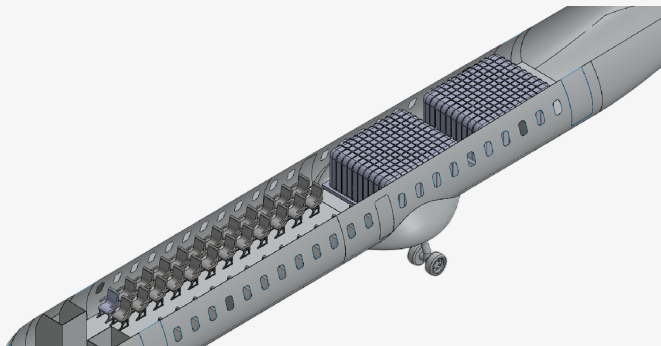


Competitive Price
USD\$1350/kg

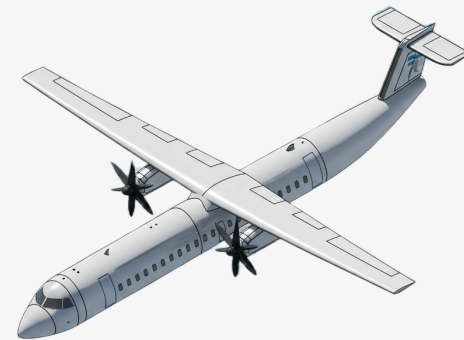
Advanced Avionics



Modular Cabin



Advanced Structure



Thank you

Questions?

